

Hindustan Copper Limited

Sub.: Expression of Interest (EoI) for Stabilizing the bad ground at (-) 120 mRL at Production Shaft of Khetri Copper Mine.

Hindustan Copper Limited (HCL) is a Government of India Enterprise under the administrative control of Ministry of Mines, Government of India incorporated on 9th November 1967, under the companies Act 1956. HCL is country's only integrated producer of refined copper from indigenous sources i.e from mining, ore beneficiation, smelting, refining and casting of refined copper metal into downstream products.

Presently HCL has five operating units across the country of which one of the important mining unit situated in State of Rajasthan known as Khetri Copper Complex. It is situated about 160 km north of State capital Jaipur and 180 km southwest of New Delhi and has two highly mechanized / one of the most modern underground mines namely Khetri & Kolihan Copper Mines.

HCL intends to enhance the production capacity of Khetri Copper Mine (KCC) from presently 0.60 Million Tonnes to 1.50 Million Tonnes per year. In view of above, apart from expansion of other lateral & vertical activities, its Production Shaft having dia of 6.10 mtrs is also being deepened from (-) 85 mRL to (-) 300 mRL.

While developing the cross cut at (-)120 mRL from the adjoining BGML shaft to come under the Production shaft location for further depth extension, a geologically disturbed and weak zone was encountered after developing the cross cut for a length of 16.5 meters, where the cross cut face had collapsed and seepage water flooded in the area. Action was taken to support the collapsed area by making steel arches and covering the roof but with further collapse steel supports erected were broken / bend.

To know the extent of weak zone, three nos. of NX size diamond drilling bore holes have also been drilled & observed that the weak strata encountered in the drilled holes is much more than what is presently exposed. The details showing the bad ground with respect to Production Shaft as indicated by the above bore holes is placed in Figure: A.

To overcome the above problems, Expressions of Interest (EoI) are invited from interested party/organization to take-up the following jobs:

01. Study the problem and suggest method of stabilizing the rock, advice the shaft wall thickness, concreting type to support the shaft for long term requirement.
02. To undertake the job of supporting for the length it extends or advice the party who could implement the suggestion as given in 01 and supervise the work suggested.

03. The situation may reoccur again in the same production shaft or the nearby service shaft being sunk. The party/organization will have long term MoU with HCL.

A write-up containing Geology, rock condition, Physico-Mechanical Properties of the rock and details of In-situ Stress determination studies by hydrofracturing method in respect of KCC mine is enclosed for reference.

Your EoI should be submitted on or before 28.03.2016 up to 3.00 pm by providing credentials to overcome similar type of jobs in the past.

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Geology of Khetri Copper Mine area

The major rock formations in Khetri Mine area are represented by metamorphosed arenaceous and argillaceous sediments with intercalated calcareous bands. These rocks are intruded by younger basic rocks and acidic intrusions of granite, pegmatite and quartz veins. The succession of major rock units found in and around Khetri mine area (from east/footwall to west/hanging wall) is as given under:-

Ajabgarh Group	5. Sericite Quartzite
	4. Phyllite/andalusite schist
	3. Garnet-chlorite-quartzite/schist+amphibole+biotite, amphibole rich rock, amphibole magnetite rock
----- Gradational Contact -----	
Alwar Group	2. Amphibole quartzite with bands of amphibole rich rock and amphibole magnetite rock
	1. Feldspathic quartzite

Amphibole quartzite, garnet chlorite quartzite/schist, amphibole rich rock and amphibole magnetite rocks are the host rocks of the main lodes / ore lenses. Feldspathic quartzite represents foot wall limit while phyllite marks the hanging wall limit of the copper mineralization. Mine has mainly following litho units:

01. Phyllite/andalusite schist:
02. Garnet chlorite quartzite/schist + amphibole + biotite, amphibole richRock, amphibole magnetite rock
03. Amphibole quartzite
04. Felspathic quartzite
05. Dolomite and impure marble
06. Intrusives

The strike of the rock varies from NE-SW to NNE – SSW with dips between 35 to 65° due NW. One important transverse fault termed as “Kudhan Fault” apart from number of criss-cross / parallel joints and small transverse faults exist in the area.

The mine has a strike length of 4.60 Kms (from 1600 to 6200 local latitudes) and divided into two blocks namely Khetri Block (1600 to 5000 latitudes) & Kolihan Block (5000 to 6200 latitude). There is continuous production from the mine since 1972 & has produced around 25.80 million tonnes so far and around 53.40 Million tonnes copper ore reserves is still available in the mines. Presently mine is producing @ 0.60 Million Tonnes copper ore per year. Expansion activities are under way to enhance the copper ore production from 0.60 Million Tonnes to 1.50 Million Tonnes per year.

While developing cross cut from BGML Shaft towards production shaft at (-) 120 mRL of the mine, up to a distance of around 16.50 meters (from centre of BGML Shaft) hard compact Quartz – Chlorite – Biotite Schist + / - Garnet rock unit was found but immediately after 16.50 mtrs crushed / weak zone had been encountered on 17.03.2015. The rock unit of the crushed zone is also Quartzite – chlorite – Biotite Schist &. Dip of the shear zone is around 43°.

The Physico-Mechanical Properties of the rock of Khetri Mine:

Sr. No.	Phyco-Mechanical Properties	Value
01.	UCS	F/w 97 MPa, H/W 160 MPa
02.	Cohesion	17.83 MPa
03.	Friction Angle	42.6 Degree
04.	Young's modulus	104 GPa
05.	Poisson's ratio	0.22
06.	PMR	65
07.	Insitu strss condition	
	a). Vertical insitu stree (Sv)	$S_v = 0.03H$ MPa
	b). Major Horizontal Insitu Stree (SH)..	$SH = 6.87 + 0.007H$ MPa
	c). Minor Horizontal Insitu Stress (Sh)	$Sh = 4.57 + 0.0051H$ Mpa
	d). major Horizontal Insitu Stress direction is along strike	

The In-situ Stress:

The National Institute of Rock Mechanics (NIRM) had conducted In-situ Stress determination studies by hydrofracturing method at Khetri Mine during 1991-92 and in 1993. The combined results of these areas indicated that the horizontal principal's stresses vary with the depth as per following equations:

$$S_H = 1.955 + 0.0327 * z$$

$$S_h = 2.557 + 0.0148 * z$$

$$S_v = 0.02943 * z$$

Where z = depth below the surface.

The major principal stress direction is given as $N25^0E$. The strike of the ore body is the north block is $N60^0E$. As the strike of the ore body is inclined to the principal stress directions, the horizontal principal stress were resolved into two components, one parallel and another perpendicular to the strike of the ore body. The variation of ore body stresses perpendicular to the strike of the ore body was evaluated as:

$$a_{XX} = 2.359 + 0.0207 * z \text{ (across the ore body)}$$

$$a_{xx} = 2.153 + 0.0268 * z \text{ (along the ore body)}$$

The vertical stress was assumed to be due to weight of the overburden strata. Its variation with depth is the same as that of S_v :

$$a_{yy} = 0.02943 * z$$

KHETRI COPPER MINE

EAST WEST SECTION AT -120 mRL

Figure - A

